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Rocky Bhatia
Roadmap
to Learn
AI SYSTEM
DESIGN
Levi's



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Why Learn AI System Design

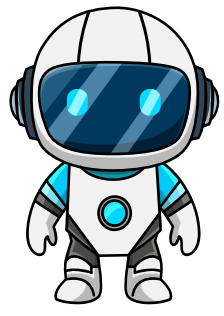
Modern AI apps are full systems, not isolated models.

They combine distributed infrastructure, data pipelines, orchestration layers, and real-time workflows.

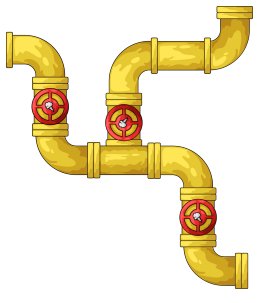
AI System Design helps you build:



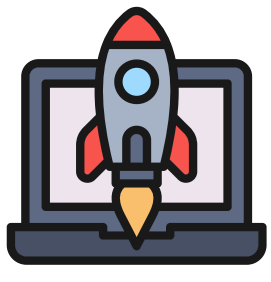
Scalable production AI apps



Enterprise-grade AI systems



Reliable ML pipelines



Real-world deployments

Focus on systems thinking over model-building.



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Step 01
Foundations First



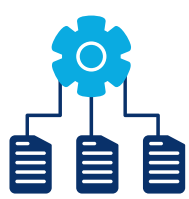
APIs, networking, and backend design



Databases (SQL vs NoSQL tradeoffs)



Caching strategies and load balancing



Distributed systems principles

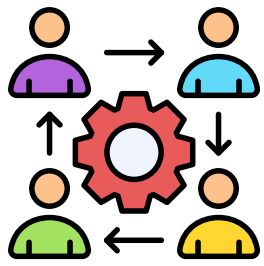
Understand bottlenecks, scaling patterns, and system reliability early



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Step 02

Core AI Concepts

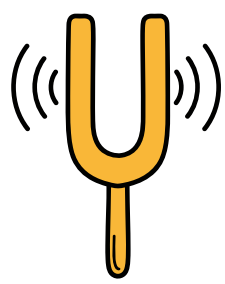


Training vs inference workflows



Embeddings and semantic representation

Learn AI from a system architecture perspective:



Fine-tuning vs prompting tradeoffs



Evaluation + benchmarking approaches

Focus on how models behave in real production systems.



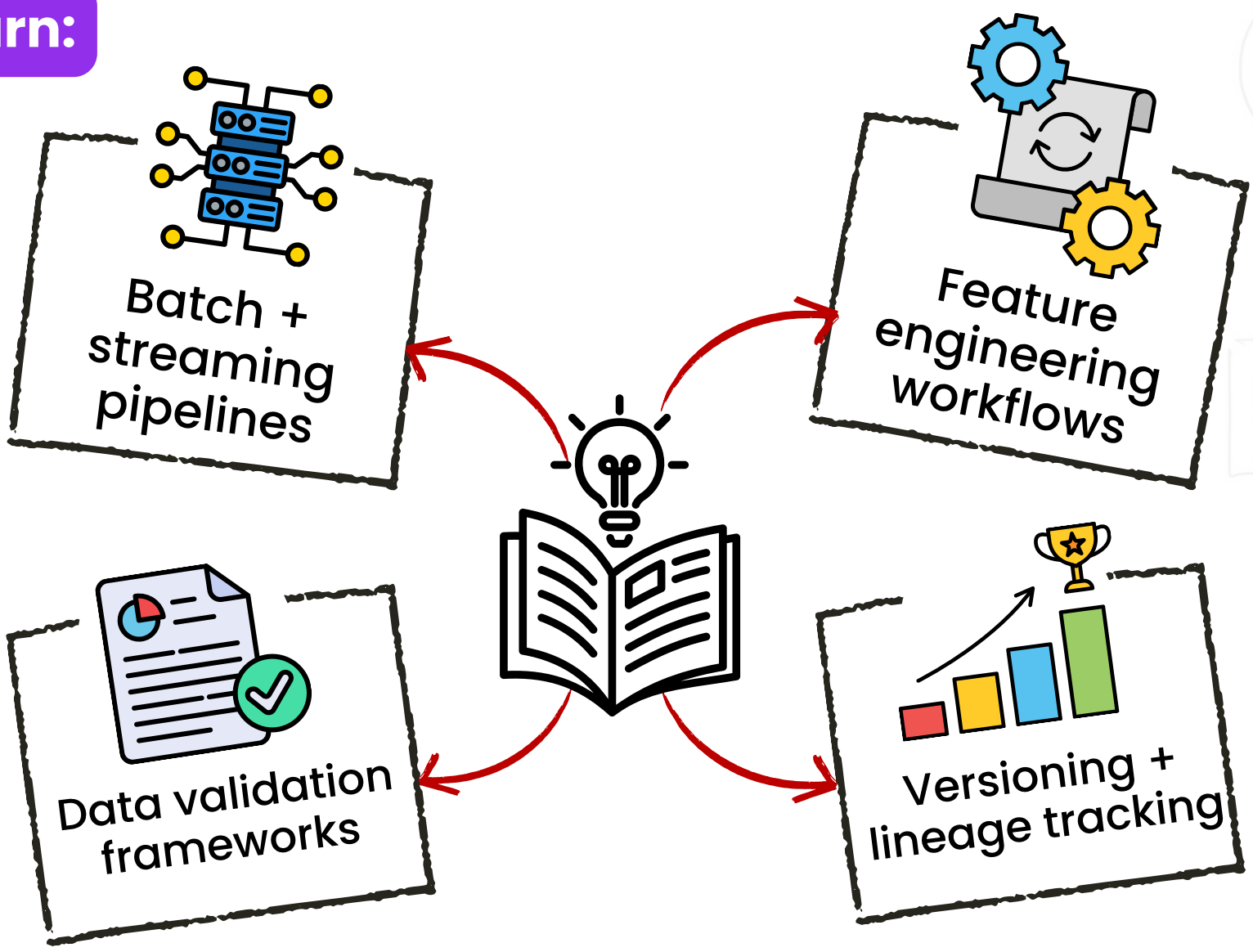
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Step 03

Data Pipeline Design

Every AI system is fundamentally a data system.

Learn:



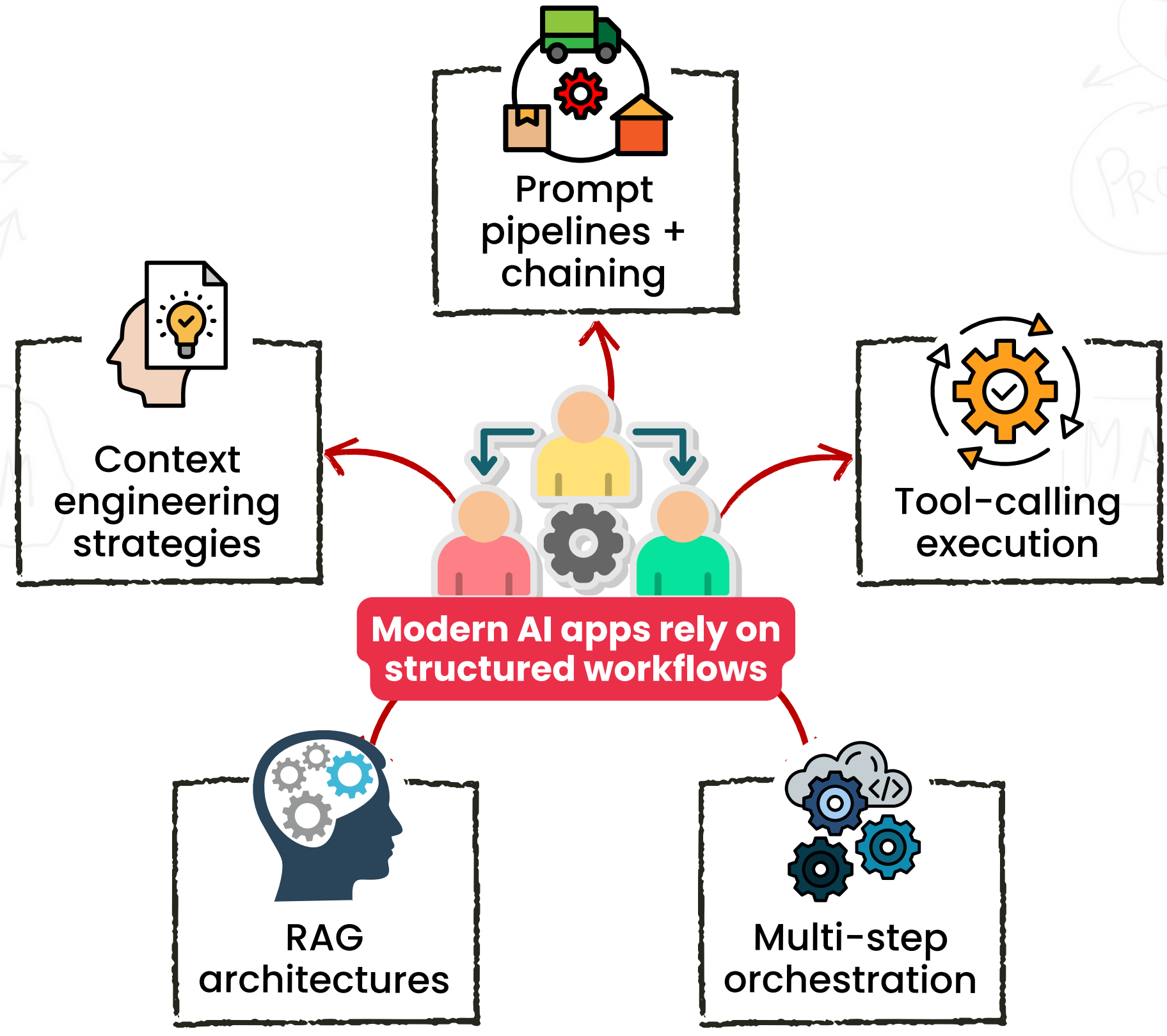
Tools: Kafka · Spark · Airflow

Understand how data quality impacts model performance.

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Step 04

LLM System Foundations



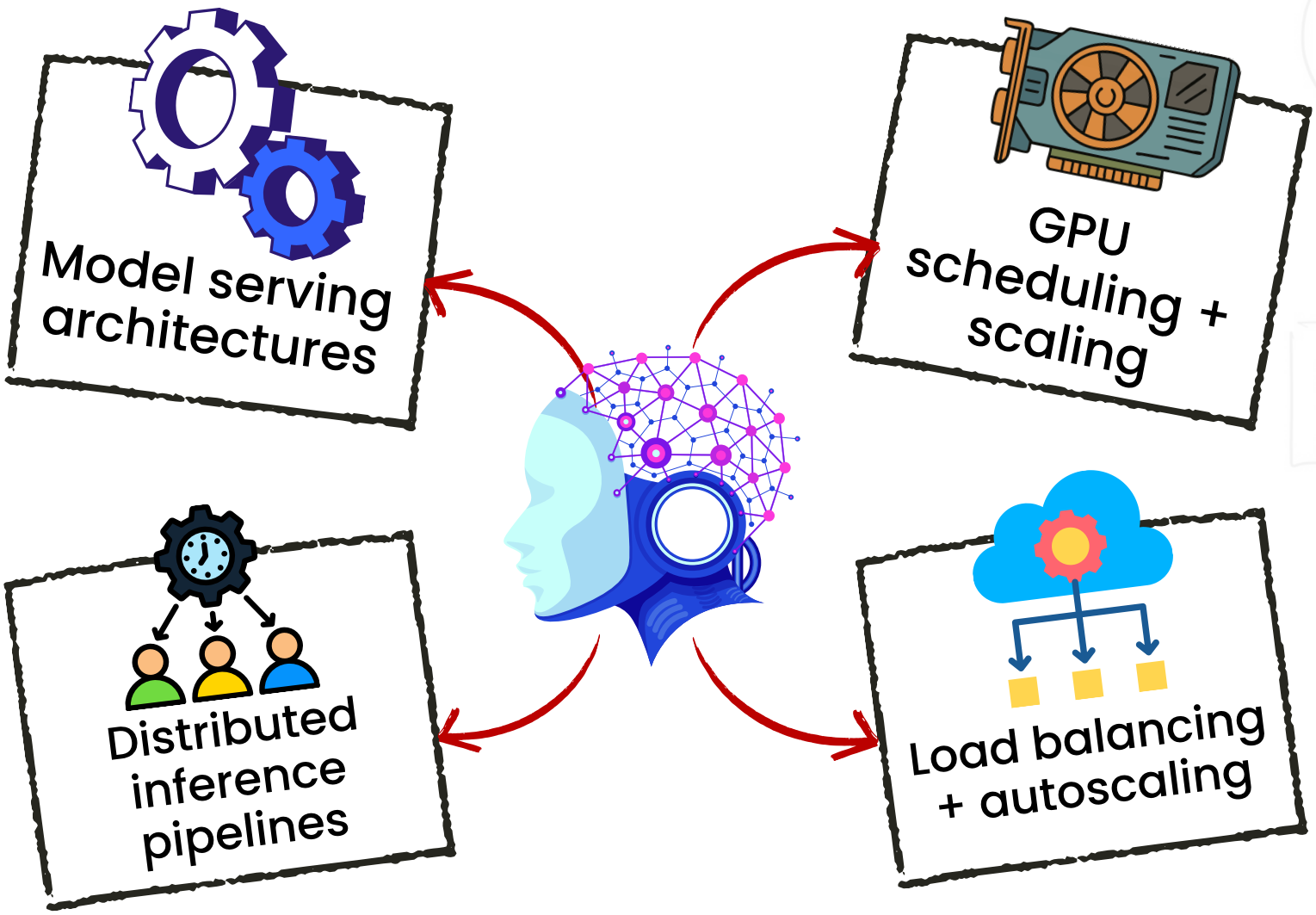
Think in workflows, not prompts.

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Step 05

AI Infrastructure

Production AI requires serious infrastructure planning:



Tools: Kubernetes Ray vLLM

Focus on performance, latency, and cost efficiency.

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Step 06

Storage + Retrieval Systems

Metadata storage strategies

Hybrid retrieval systems

Vector databases

AI systems rely heavily on retrieval design:

Index optimization

Caching layers

Tools: Pinecone FAISS Redis

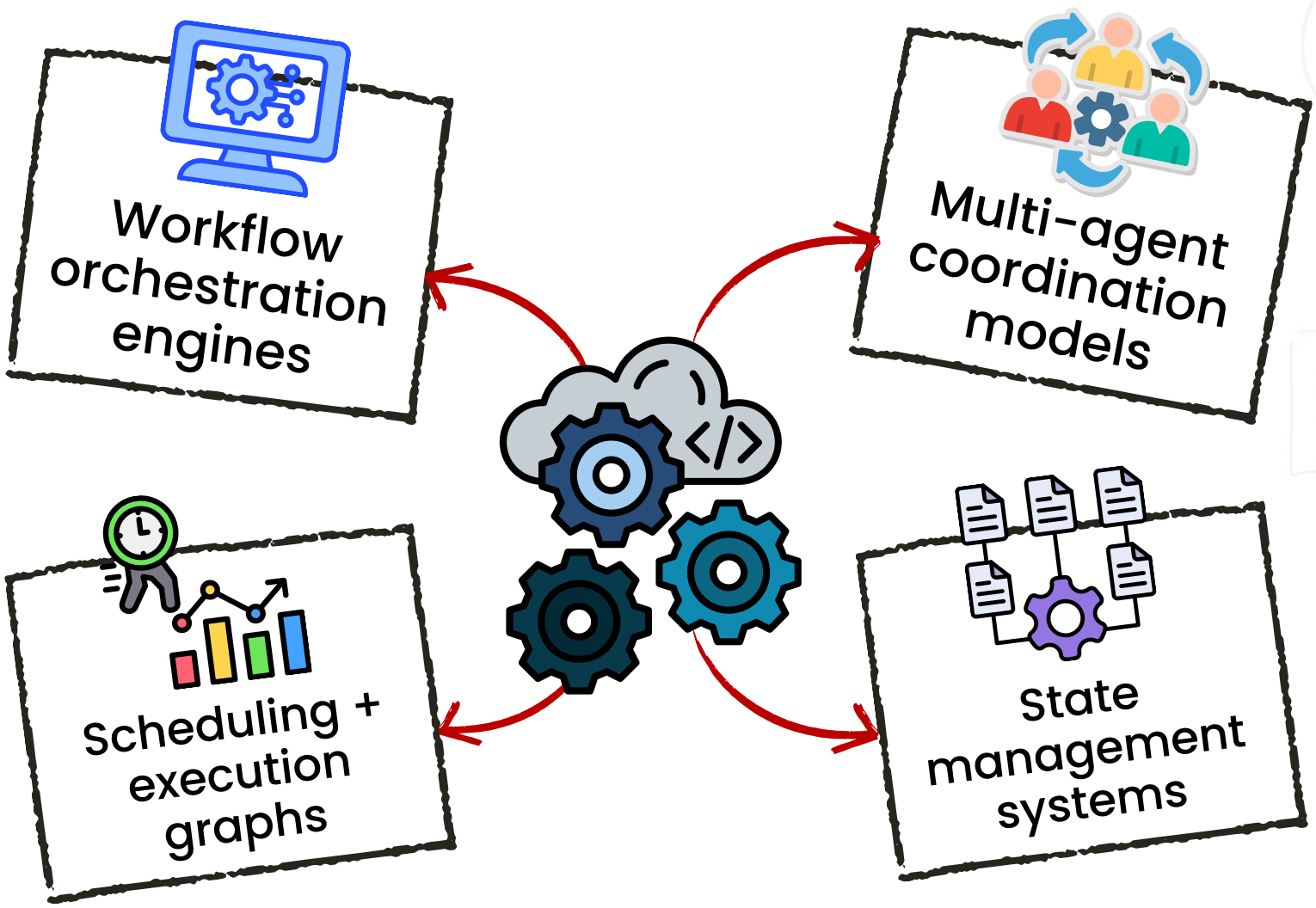
Design for low-latency retrieval at scale.

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Step 07

AI Orchestration

Complex AI systems require coordination layers:



Tools: LangGraph CrewAI Temporal

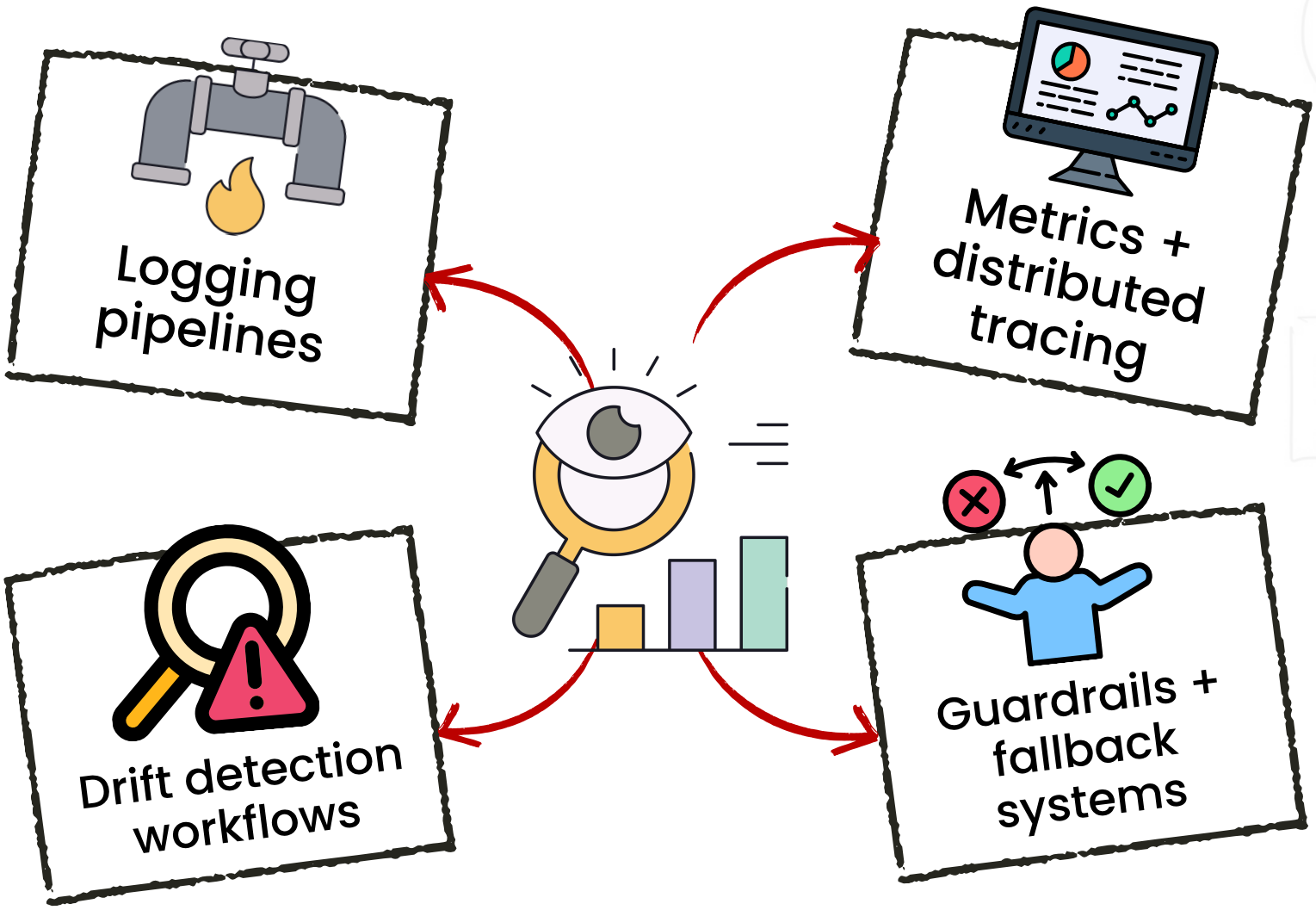
Focus on reliability and execution control.

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Step 08

Observability + Reliability

Production AI must be observable and stable:



Tools: Prometheus Grafana

Track performance across every system layer.

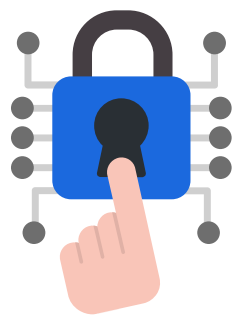
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Step 09

Security + Governance



Data privacy + isolation systems

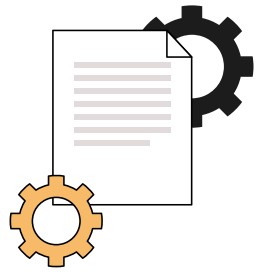


Access control frameworks

Enterprise AI systems require strong controls:



Prompt injection protection



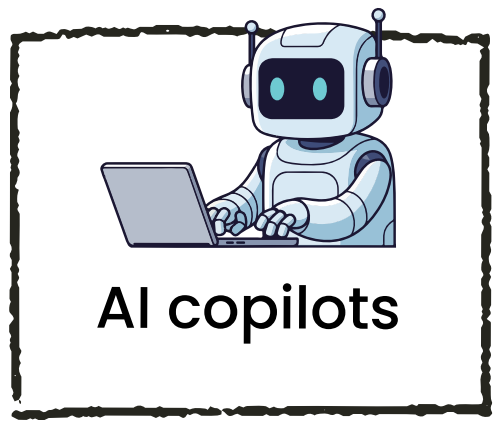
Audit + compliance workflows

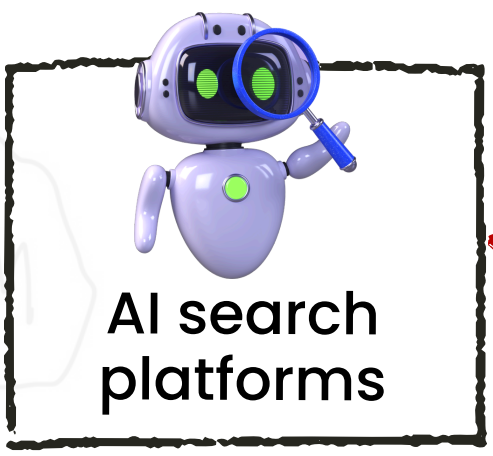
Design for trust, safety, and regulation from day one.

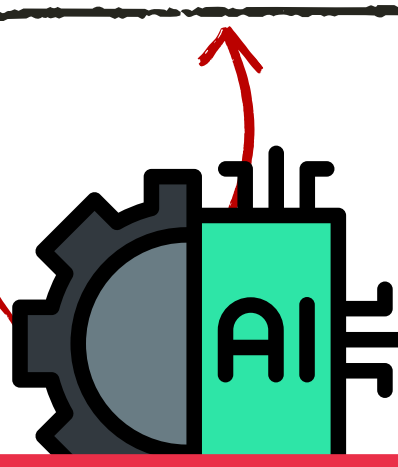
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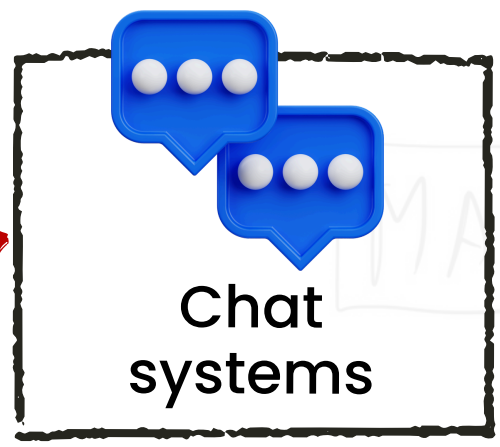
Step 10

Real System Practice









Practice designing real-world AI systems





Always evaluate tradeoffs during design.



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Step 11

Learn by Reverse Engineering

ChatGPT-style systems

GitHub Copilot workflows

Study real-world architectures deeply:

Perplexity pipelines

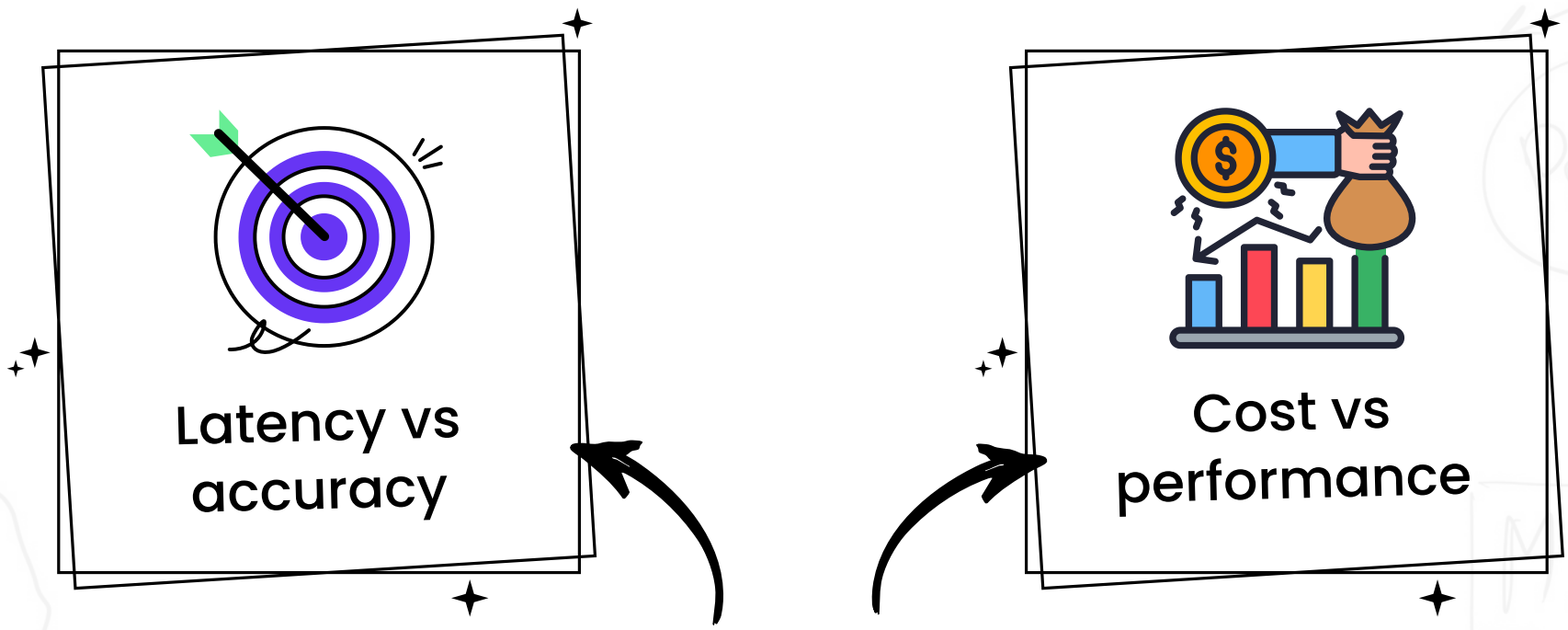
Enterprise RAG systems

Learn reusable design patterns from real implementations.

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Step 12

Master Tradeoffs



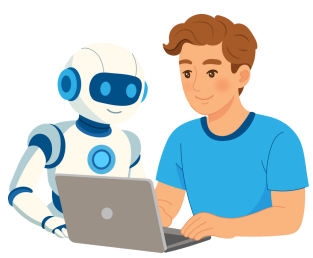
Always design intentionally with constraints in mind.

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AI System Design


Distributed systems

+


ML

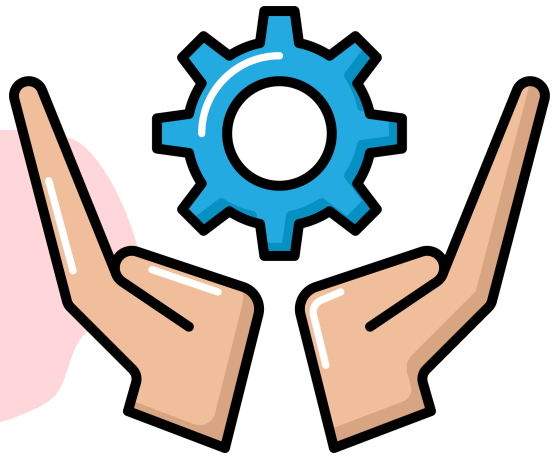
+


infrastructure

+


workflows

Mastering these together helps you design scalable, reliable, production-ready AI systems confidently.





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